

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____

Identify the parent family and state domain/range

Example (Identify the parent family and state domain/range):

$f(x) = |x| - 2$: parent family is **absolute value**; domain is all reals; range is $y \geq -2$.

Directions: Name the parent family, then give the domain and range.

1. $f(x) = x^3 - 1$

3. $f(x) = \frac{1}{x-2}$

2. $f(x) = \sqrt{x+4}$

State max/min, vertex, and intervals

Example (State max/min, vertex, and intervals):

$y = (x+2)^2 - 5$: vertex $(-2, -5)$; opens up, minimum; decreasing $x < -2$, increasing $x > -2$.

Directions: State max or min, the vertex, and both intervals.

4. $y = -(x-3)^2 + 2$

5. $y = (x+1)^2 - 4$

Determine even, odd, or neither

Example (Determine even, odd, or neither):

$f(x) = x^4$: $f(-x) = (-x)^4 = x^4 = f(x)$, so it's **even**.

Directions: Substitute $-x$, simplify, and classify.

6. $f(x) = x^3 - x$

8. (challenge) $f(x) = x^3 + 4$

7. $f(x) = x^2 + 3$

Challenge (same sheet):

9. **(challenge)** Write a function of your own that is a shifted square-root function, then find its domain, range, and classify its symmetry.

Answer Key — fully worked

Procedure: Step 1 — Name the parent family from the equation's shape.; Step 2 — For quadratics, find the vertex and determine max/min and intervals.; Step 3 — Check for domain restrictions (denominators, radicals) and determine the range.; Step 4 — Substitute $-x$ to test even, odd, or neither..

1. Cubic family; domain all real numbers; range all real numbers
2. Square root family; domain $x \geq -4$; range $y \geq 0$
3. Reciprocal family; domain $x \neq 2$; range $y \neq 0$
4. Opens down, maximum; vertex $(3, 2)$; increasing $x < 3$, decreasing $x > 3$
5. Opens up, minimum; vertex $(-1, -4)$; decreasing $x < -1$, increasing $x > -1$
6. $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$: odd
7. $f(-x) = x^2 + 3 = f(x)$: even
8. $f(-x) = -x^3 + 4$, matching neither $f(x)$ nor $-f(x)$: neither
9. Answers vary; a sample valid response is $f(x) = \sqrt{x-2} + 3$. Domain: the radicand must be ≥ 0 , so $x - 2 \geq 0$, giving $x \geq 2$. Range: the smallest output is at $x = 2$, where $f(2) = \sqrt{0} + 3 = 3$, and the graph only rises from there, so $y \geq 3$. Symmetry: the domain $x \geq 2$ is not symmetric about the origin (e.g., $x = 2$ is allowed but $x = -2$ is not), so the function is neither even nor odd. Any correctly analyzed shifted square-root function earns full credit.